TECHNISCHE UNIVERSITÄT MÜNCHEN

Quantum Chemistry Laboratory

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Full Solution of Exercise 7

a. **Orbital energies**

energy [E_h]	symmetry	assignment
-25.90	σ	F-1s
-1.47	σ	F-2s
-0.59	σ	HF-bonding
-0.46	π	F-2p_x
-0.46	π	F-2p_y

b. The **first IP** amounts to 0.46 E_h (12.63 eV, experimental value: 16.05 eV). Improvement by carrying out another calculation for the ion and subtracting the total energies possible.

E(HF ion, ground state) - E(HF neutral, ground state) = IP

c. The **IP of the core orbital** (Koopmans) is 705 eV = 68.0 MJ/mol = 25.9 hartree. This energy corresponds to a photon of the wavelength 1.76 nm. The radiation belongs to the range of *X*-rays.

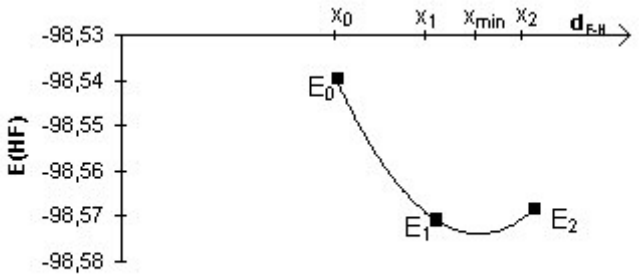
d. Energies in E_h

sum of occupied orbitals:	-57.770
total energy:	-98.571
core repulsion:	5.194

electronic Hartree-Fock energy: -103.765

The difference of 45.995 E_h is caused by the double count of coulombic and exchange terms when adding the orbital energies.

Supplementary Exercise



Distance [Å]	Total energy [E_h]
0.817	-98.539818
0.917	-98.570780
1.017	-98.568489

a. With $u = x - x_1$, the parabola is:

$f(u) = au^2 + bu + d = 1.66265\ u^2 - 0.143355\ u + d$

One obtains $x_{min} = 0.9601$ Å.

b. The potential curve of a harmonic oscillator is $V = 1/2\ kx^2$. Since the second derivatives of the two potentials are equal, one obtains:

$k = 2a = 3.33253\ E_h/Å^2 = 1449.7\ J/m^2$

and

$\omega = (k/\mu)^{1/2} = 9.551 \cdot 10^{14}\ s^{-1}$

using the reduced mass $\mu = 1.58924 \cdot 10^{-27}$ kg. The wave number of the harmonic vibration is therefore:

$\bar{\nu} = \frac{\omega}{\ } = 5070\ cm^{-1}$

$$\nu = \frac{}{2\pi c} =$$



[General help]



[To overview]



[Compute]



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